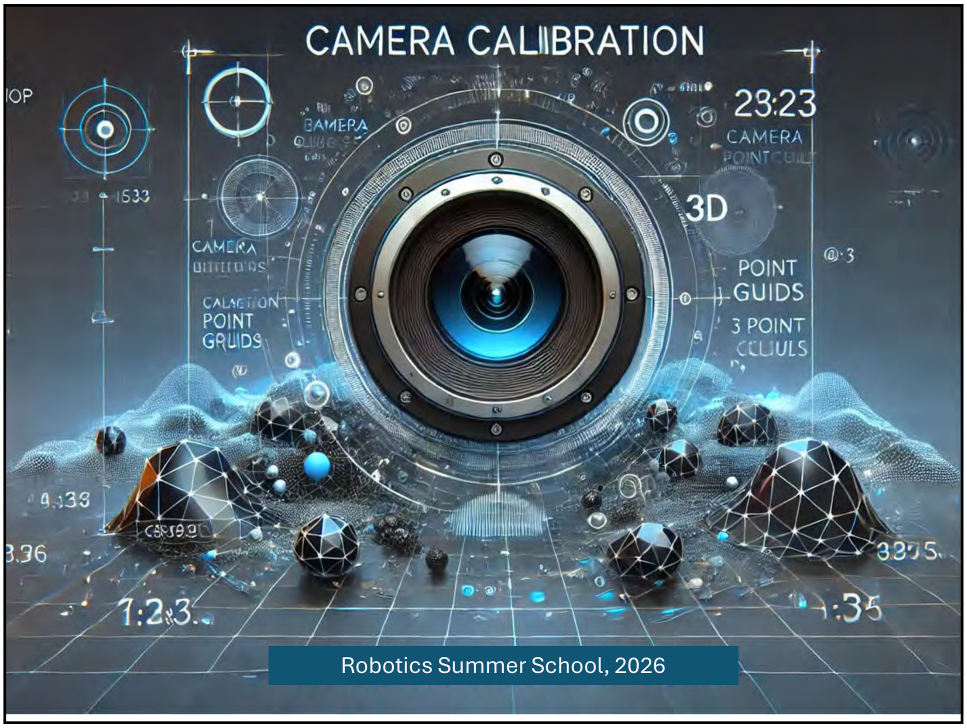


Robotics Summer School, 2026

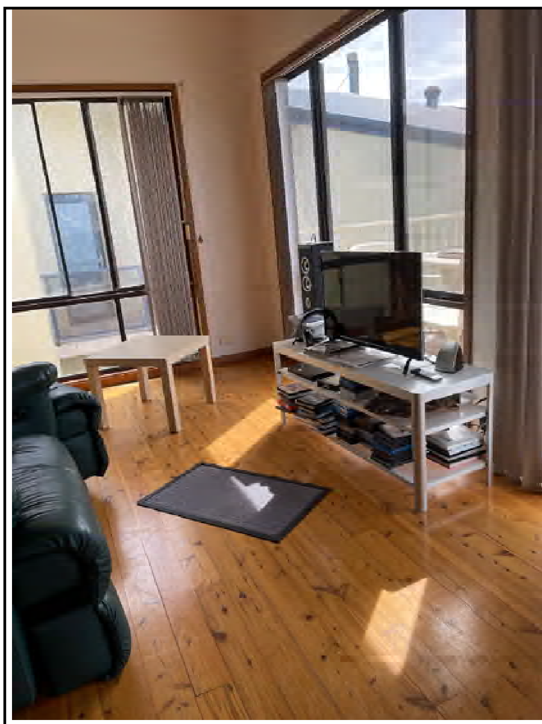
1

- What is the field of view of the image (in degrees)
- Is the camera horizontal / tilting up / down?
- By how many degrees?
- What direction is the camera facing with respect to the room orientation?
- What direction is the TV facing?
- What is the angle of slope of my ceiling?

2

- What is the field of view of the image (in degrees)
- Is the camera horizontal / tilting up / down?
- By how many degrees?
- What direction is the camera facing with respect to the room orientation?
- What direction is the TV facing?
- What is the angle of slope of my ceiling?
- What direction is the sun with respect to the orientation of the room?
- What is the elevation of the sun?
- What direction does my living room face, given that this picture was taken at 9:58am.
- What is the latitude / longitude of my house?

3



- What is the field of view of the image (in degrees)
- Is the camera horizontal / tilting up / down?
- By how many degrees?
- What direction is the camera facing with respect to the room orientation?
- What is the angle of slope of my ceiling?
- What direction is the TV facing?
- What direction is the sun with respect to the orientation of the room?
- What is the elevation of the sun?
- What direction does my living room face, given that this picture was taken at 9:58am.
- What is the latitude / longitude of my house?
- I have rotated the TV table. By how many degrees?
- What is the orientation of the black mat?

4

Planar robot motion: the camera has moved and rotated (about the vertical). How many degrees has it rotated?

5

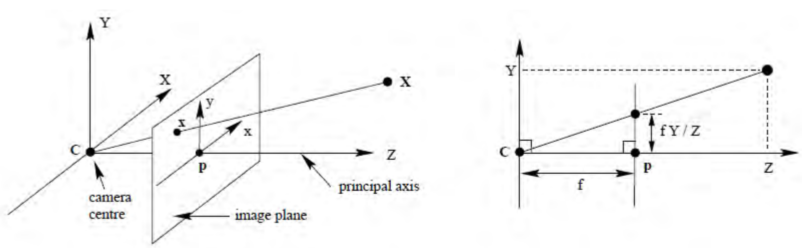
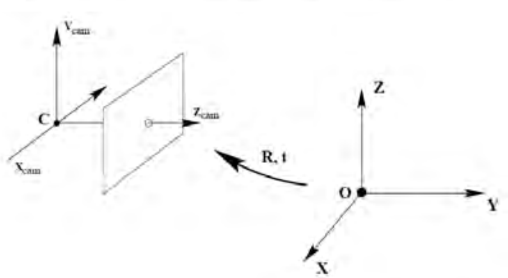


Fig. 6.1. **Pinhole camera geometry.** C *is the camera centre and* p *the principal point. The camera centre is here placed at the coordinate origin. Note the image plane is placed in front of the camera centre.*



6

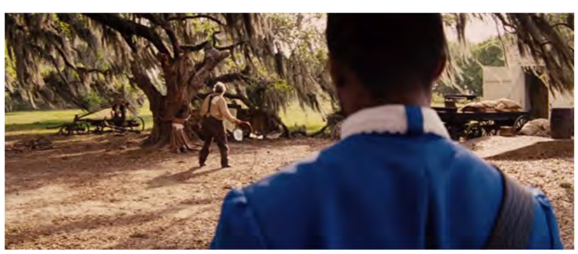
$$\mathrm{x} = \mathrm{KR}[\mathrm{I} \mid -\tilde{\mathrm{C}}]\mathrm{X}$$

$$\mathrm{K} = \begin{bmatrix} \alpha_x & & x_0 \\ & \alpha_y & y_0 \\ & & 1 \end{bmatrix} \quad \mathrm{K} = \begin{bmatrix} \alpha_x & s & x_0 \\ & \alpha_y & y_0 \\ & & 1 \end{bmatrix}.$$

7

What does camera calibration give?

1. Proper "perspective".
   without proper understanding, scene is distorted.



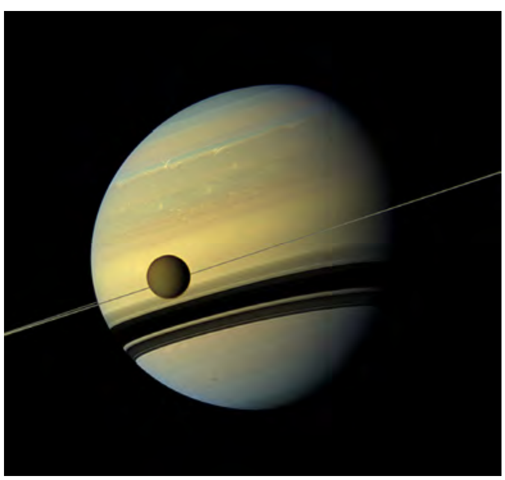
Django frame.

Close-up and wide-angle images emphasize differences in size.

8

Sometimes it is best to use
long-focal length images.

9



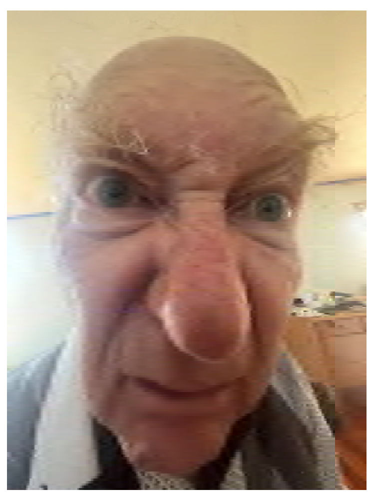
Long focal length image.

10

For pictures of humans,
It is best to use long
focal length to avoid
perspective distortion.

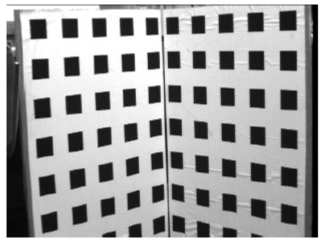
11

For pictures of humans,
It is best to use long
focal length to avoid
perspective distortion.



12

History.



Calibration grids.

13

My view:

Accurate calibration is often not necessary.

In most cases, the scene provides its own
Camera calibration.

Even in SfM.

14

Quick and dirty camera calibration

15

Quick calibration

$$\boxed{f = \frac{zp}{s}}$$

- f = focal-length parameter.
- z = camera distance
- s = reference length (mm)
- p = reference image length (pixels)

	A	B	C	D
1		**Width**	**Height**	**Ratio**
2	Image			
3	W/H (pixels)	2016.00	1512.00	1.333
4	W/H (mm)	1162.00	883.00	1.316
5	Pixels / mm	1.73	1.71	
6	Camera distance (mm)	910.00	910.00	
7	f (pixels)	1578.80	1558.23	1.013
8				
9	Tan (theta_h)	0.638	0.485	
10	theta/2	32.557	25.881	
11				
12	Width start (mm)	66.00	85.00	
13	width end (mm)	1228.00	968.00	

16

Quick calibration

$$f = \frac{zp}{s}$$

- f = focal-length parameter.
- z = camera distance
- s = reference length (mm)
- p = reference image length (pixels)

	A	B	C	D
1		Width	Height	Ratio
2	Image			
3	W/H (pixels)	2016.00	1512.00	1.333
4	W/H (mm)	1162.00	883.00	1.316
5	Pixels / mm	1.73	1.71	
6	Camera distance (mm)	910.00	910.00	
7	f (pixels)	1578.80	1558.23	1.013
8				
9	Tan (theta_h)	0.638	0.485	
10	theta/2	32.557	25.881	
11				
12	Width start (mm)	66.00	85.00	
13	width end (mm)	1228.00	968.00	

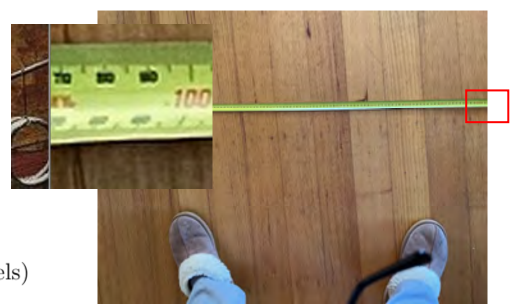
17

Quick calibration

$$f = \frac{zp}{s}$$

- f = focal-length parameter.
- z = camera distance
- s = reference length (mm)
- p = reference image length (pixels)

	A	B	C	D
1		Width	Height	Ratio
2	Image			
3	W/H (pixels)	2016.00	1512.00	1.333
4	W/H (mm)	1162.00	883.00	1.316
5	Pixels / mm	1.73	1.71	
6	Camera distance (mm)	910.00	910.00	
7	f (pixels)	1578.80	1558.23	1.013
8				
9	Tan (theta_h)	0.638	0.485	
10	theta/2	32.557	25.881	
11				
12	Width start (mm)	66.00	85.00	
13	width end (mm)	1228.00	968.00	



18

Quick calibration

$$f = \frac{zp}{s}$$

- f = focal-length parameter.
- z = camera distance
- s = reference length (mm)
- p = reference image length (pixels)

	A	B	C	D
1		**Width**	**Height**	**Ratio**
2	Image			
3	W/H (pixels)	2016.00	1512.00	1.333
4	W/H (mm)	1162.00	883.00	1.316
5	Pixels / mm	1.73	1.71	
6	Camera distance (mm)	910.00	910.00	
7	f (pixels)	1578.80	1558.23	1.013
8				
9	Tan (theta_h)	0.638	0.485	
10	theta/2	32.557	25.881	
11				
12	Width start (mm)	66.00	85.00	
13	width end (mm)	1228.00	968.00	

19

Quick calibration

$$f = \frac{zp}{s}$$

- f = focal-length parameter.
- z = camera distance
- s = reference length (mm)
- p = reference image length (pixels)

	A	B	C	D
1		**Width**	**Height**	**Ratio**
2	Image			
3	W/H (pixels)	2016.00	1512.00	1.333
4	W/H (mm)	1162.00	883.00	1.316
5	Pixels / mm	1.73	1.71	
6	Camera distance (mm)	910.00	910.00	
7	f (pixels)	1578.80	1558.23	1.013
8				
9	Tan (theta_h)	0.638	0.485	
10	theta/2	32.557	25.881	
11				
12	Width start (mm)	66.00	85.00	
13	width end (mm)	1228.00	968.00	

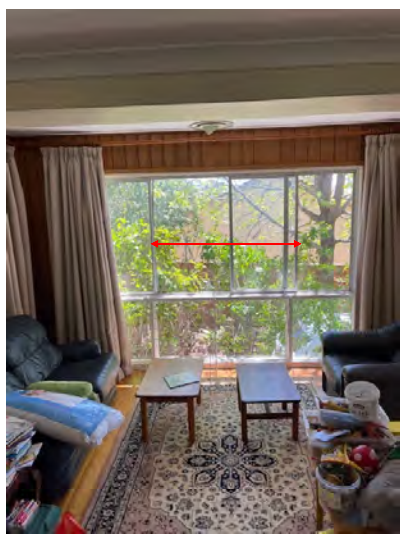
20

	A	B	C	D	E	F	G	H	I
1		**Width**	**Height**	**Diag**	**Ratio**				
2	Image								
3	W/H (pixels)	2016.00	1512.00	2344.31	1.333				
4	W/H (mm)	1442.00	1058.00	1660.00	1.363				
5	Pixels / mm	1.40	1.43	1.41					
6	Camera distance (mm)	910.00	910.00	910.00					
7	f (pixels)	1272.23	1300.49	1285.13		**Computation of diagonal (pixels)**			
8	Aspect	1.000	1.022	1.010		1508	116		
9						132	2014		
10	Tan (theta_h)	0.792	0.581	0.912		1376	-1898	2344.308	
11	theta/2	38.390	30.170	42.368					
12									
13	Width start (mm)	231.00	522.00	130.00					
14	width end (mm)	1673.00	1580.00	1790.00					

Radial distortion computation

21

Far view



Near view

	Far view	Near view
Image W (pixels)	640	640
s (pixels)	241	177
s (mm)	1500	1500
distance (mm)	7748	4220
f	1245	498
Zoom	2.5	1.0

22

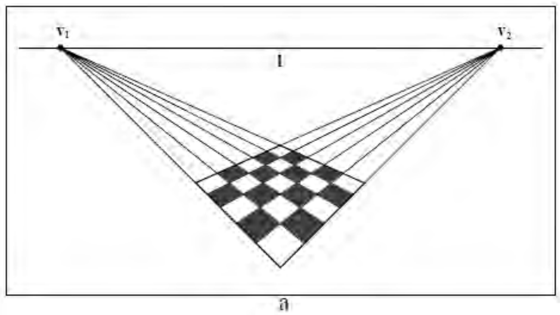
Car 1	Car 2	
2016	2016	Image W (pixels)
417	209	s (pixels)
1200	1400	s (mm)
4514	10507	distance (mm)
1569	1569	f
1	1	Zoom

23

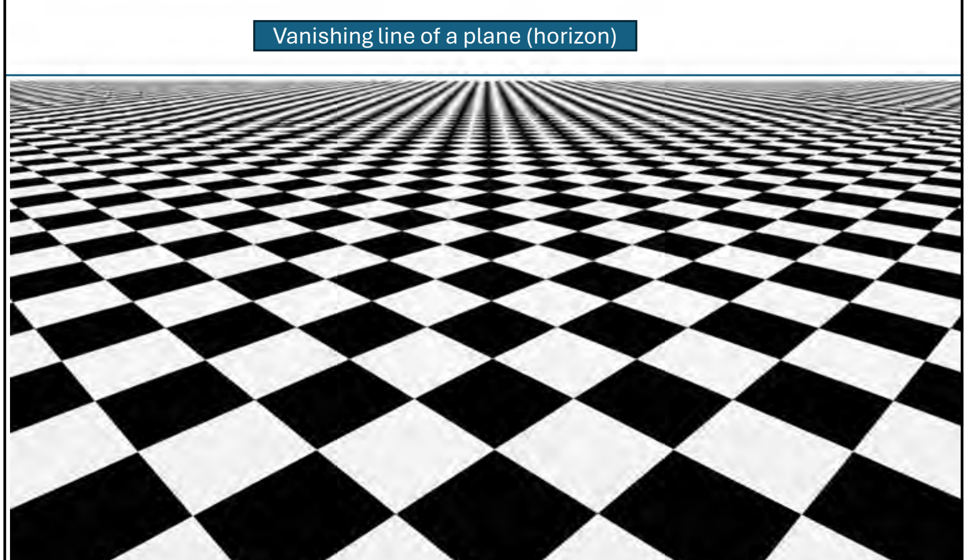
Clues to calibration are (almost) everywhere.

24

Calibration from Regular patterns

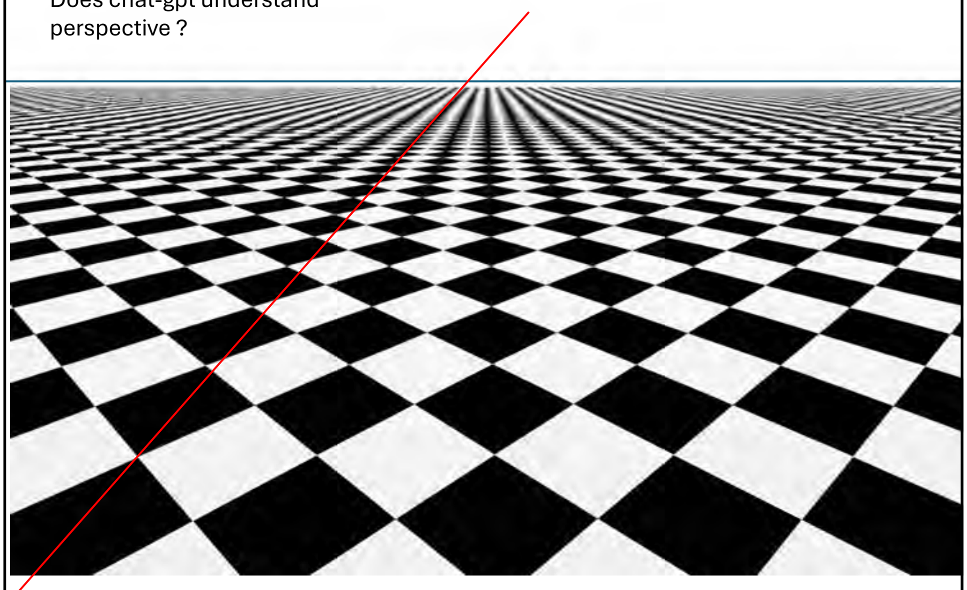


25

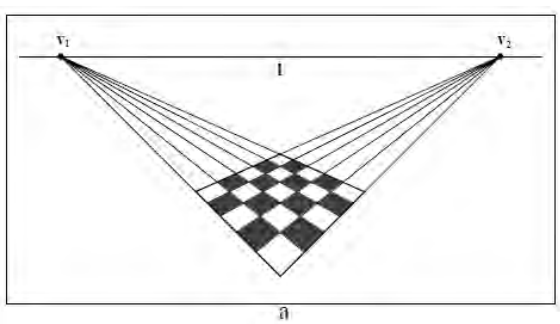


26

Does chat-gpt understand perspective ?



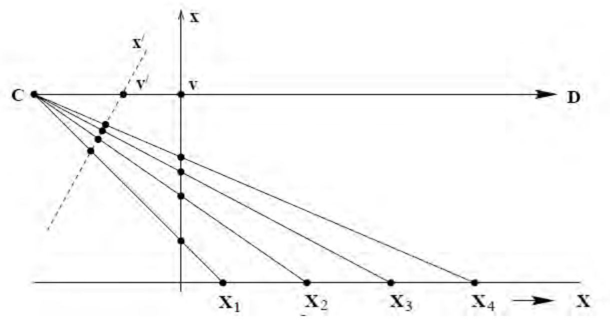
27



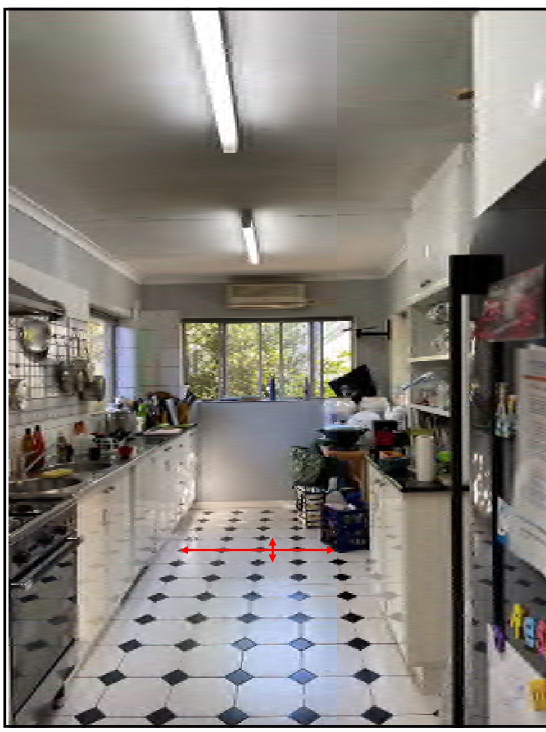
Constructing the horizon. Find two pairs of parallel lines.

28

Foreshortening



29



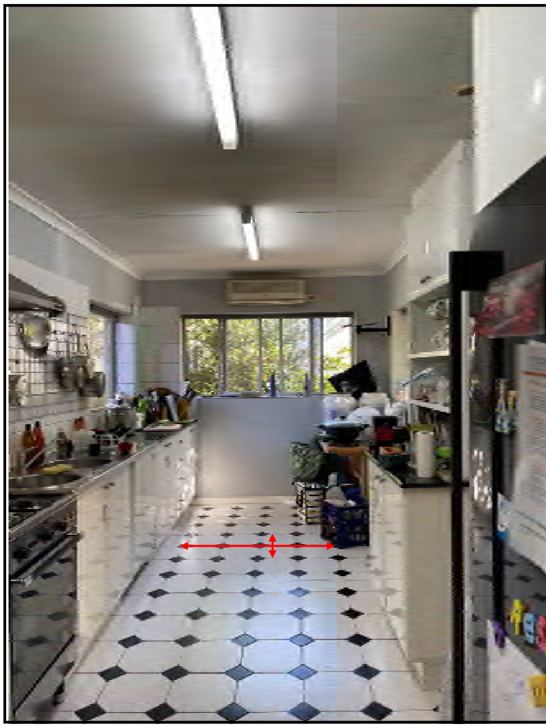
Calibration through foreshortening (simple case)

$$\alpha = \sin(\theta)$$
$$f = p \cot \theta$$

- α = foreshortening
- θ = declination angle
- p = declination in pixels

Image dim	4032	
d	3033	
p	1017	
width	838	4
height	67.5	
aspect ratio	0.32	
theta	0.33	18.80
	2988.14	

30



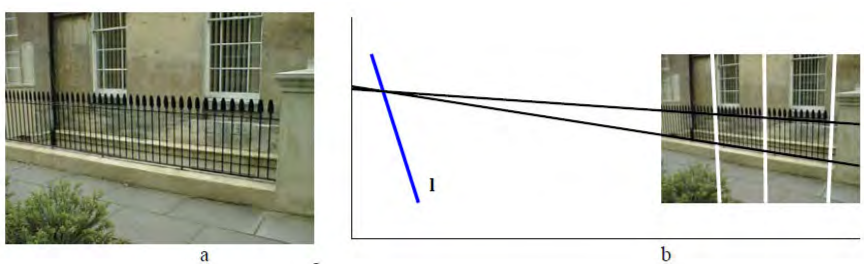
Calibration through foreshortening (simple case)

$$f = \frac{p\sqrt{1-\alpha^2}}{\alpha}$$

- f = focal length parameter
- α = foreshortening ratio
- p = pixel distance from centre of image

Image dim	4032	
d	3033	
p	1017	
width	838	4
height	67.5	
aspect ratio	0.32	
theta	0.33	18.80
	$\boxed{2988.14}$	

31



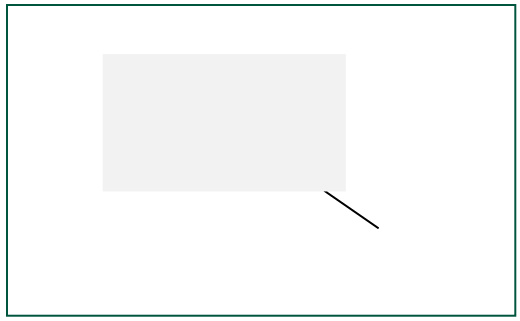
Evenly spaced parallel lines

32

33

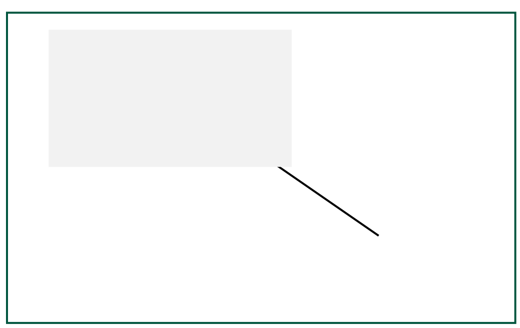
Vanishing points and lines

34



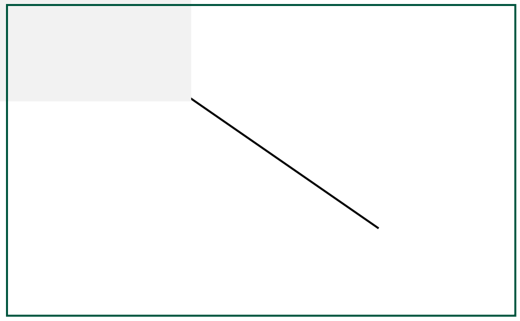
A line seen in an image.

35



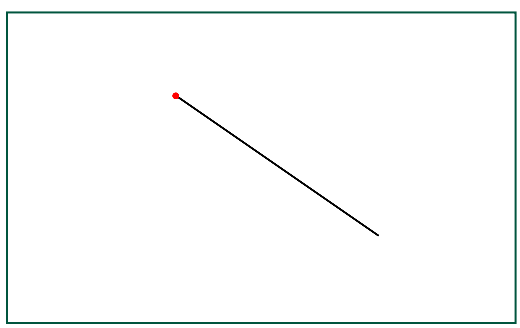
A line seen in an image - extend.

36



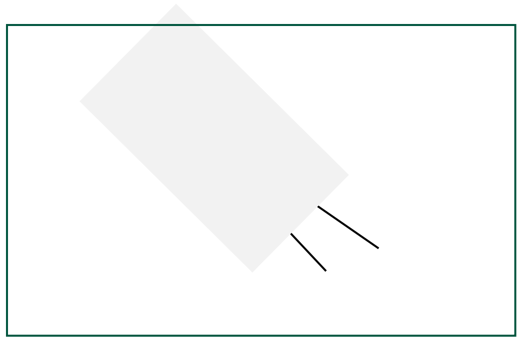
A line seen in an image – keep extending.

37



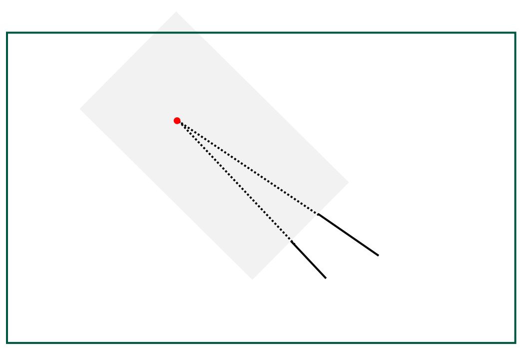
A line seen in an image – extends to a vanishing point .

38



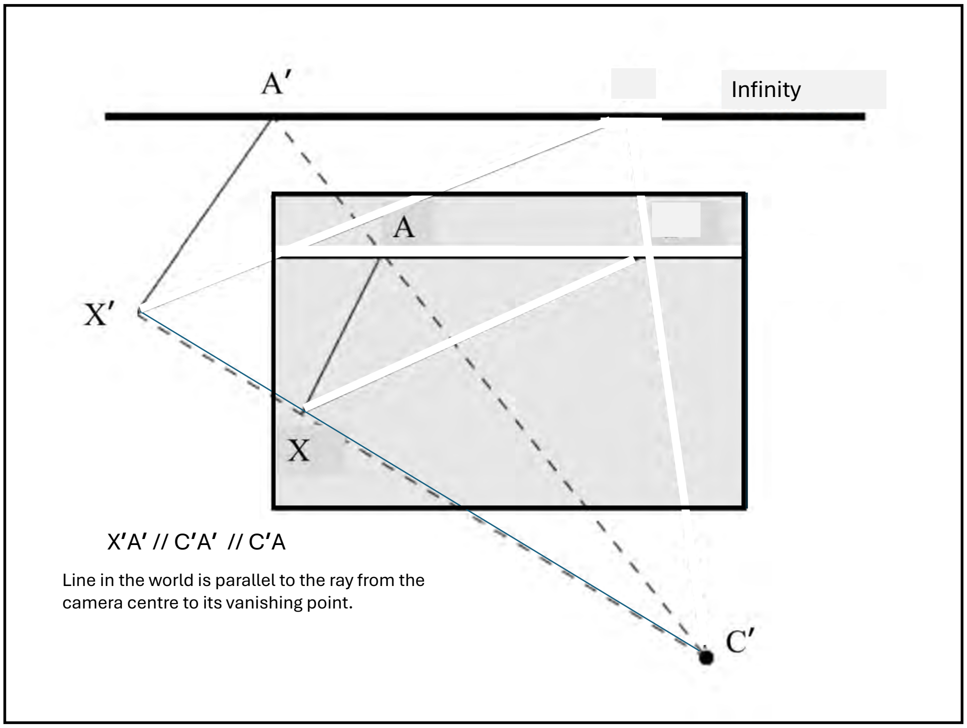
Parallel lines in the image. Vanishing points of parallel lines are the same.

39

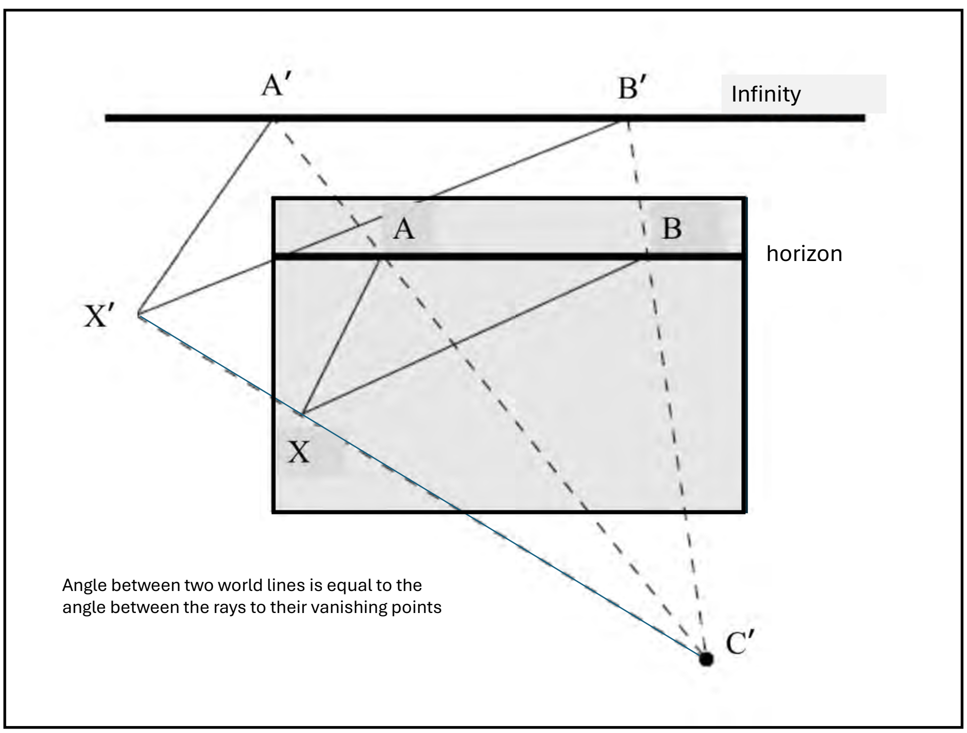


Extend parallel lines until they meet at the vanishing point.

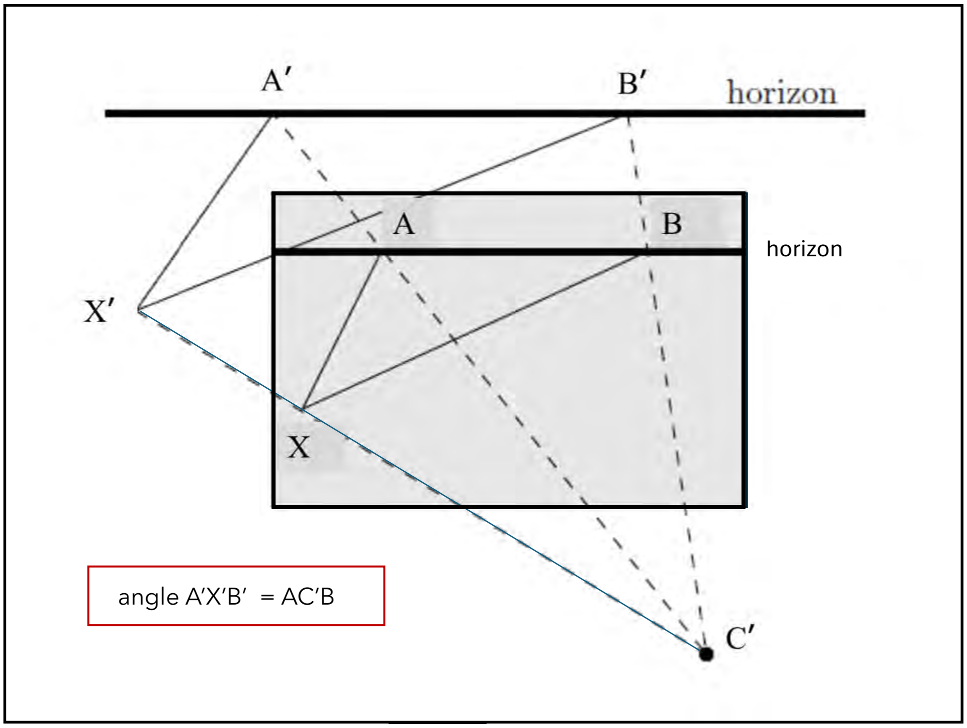
40


A′
Infinity
A
X′
X
C′
X′A′ // C′A′ // C′A
Line in the world is parallel to the ray from the camera centre to its vanishing point.

41


A′
B′
Infinity
A
B
horizon
X′
X
C′
Angle between two world lines is equal to the angle between the rays to their vanishing points

42



43

Angle between two rays.

$$\omega = (\mathtt{KK}^{\top})^{-1},$$

Image of the Absolute conic (IAC)

$$\cos(\theta) = \frac{\mathbf{x}_1{}^{\top}\omega\mathbf{x}_2}{\sqrt{\mathbf{x}_1{}^{\top}\omega\mathbf{x}_1}\sqrt{\mathbf{x}_1{}^{\top}\omega\mathbf{x}_1}} .$$

In particular, if the two rays are orthogonal, then

$$\mathbf{x}_1{}^{\top}\omega\mathbf{x}_2 = 0 .$$

Problem solved?

44

45

46

47

- Vanishing point for any line parallel with the ground plane will lie on the horizon (whether line lies on the ground plane of not).
- Vanishing points of all parallel lines will be the same.

48

- Vanishing point for any line parallel with the ground plane will lie on the horizon (whether line lies on the ground plane of not).
- Vanishing points of all parallel lines will be the same.

Room probably square: Vanishing points of tops of windows are orthogonal.

49

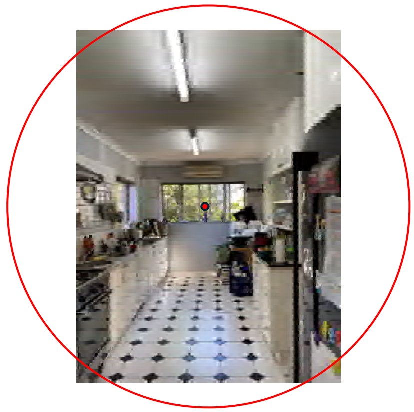
Visualizing calibration. The Calibrating conic.

Set of points in the image that lie at 45-degrees from the image centre (principal point).

A visual way to show the camera calibration.

50

The calibrating conic



51

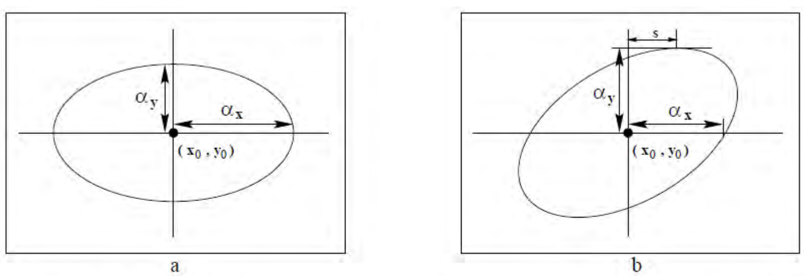


Fig. 8.27. **Reading the internal camera parameters K from the calibrating conic.** *(a) Skew s is zero. (b) Skew s is non-zero. The skew parameter of* K *(see (6.10–p157), is given by the x-coordinate of the highest point of the conic.*

$$C = K^{-T}\begin{bmatrix} 1 & & \\ & 1 & \\ & & -1 \end{bmatrix}K^{-1}$$

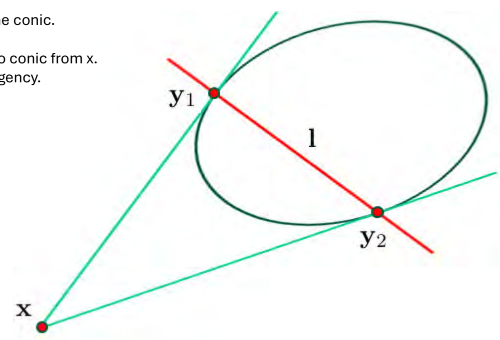
52

Geometry interlude – maybe interesting, but inessential.

Polar of a point with respect to a conic.

Case 1: point outside the conic.

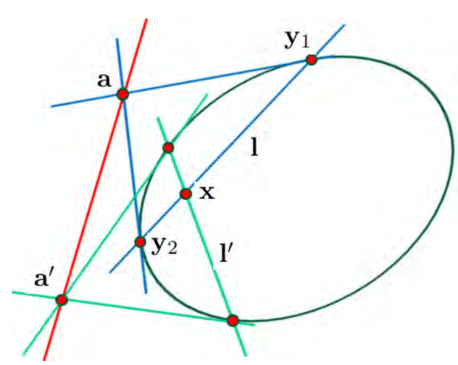
- Construct tangents to conic from x.
- Join the points of tangency.



53

Polar of a point with respect to a conic.

Case 1: point x inside the conic.
Slightly more complex geometric construction.



54

Calibrating conic allows us to find all rays (image point) perpendicular to a given direction.

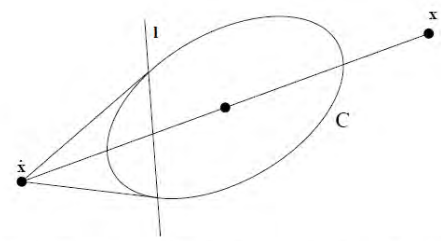
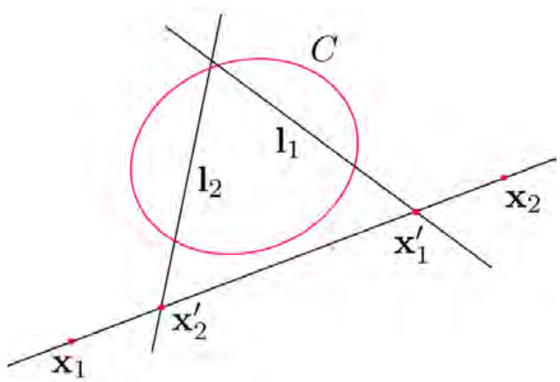


Fig. 8.28. *To construct the line perpendicular to the ray through image point* $\mathbf{x}$ *proceed as follows: (i) Reflect* $\mathbf{x}$ *through the centre of* $\mathtt{C}$ *to get point* $\dot{\mathbf{x}}$ *(i.e. at the same distance from the centre as* $\mathbf{x}$*). (ii) The desired line is the polar of* $\dot{\mathbf{x}}$.

Reflected polar construction. All points (rays) on line l are perpendicular to ray x.

55



Theorem 2.2. *Let* $\mathbf{x}_1$ *and* $\mathbf{x}_2$ *be two points and* $\mathtt{C}$ *the calibrating conic. Denote by* $\mathbf{l}_1 = \mathtt{C}\dot{\mathbf{x}}_1$ *and* $\mathbf{l}_2 = \mathtt{C}\dot{\mathbf{x}}_2$*, their reflected polars. Construct the line through* $\mathbf{x}_1$ *and* $\mathbf{x}_2$*; call it* $\mathbf{x}_1 \times \mathbf{x}_2$*. Let* $\mathbf{x}'_1$ *be the intersection of* $\mathbf{x}_1 \times \mathbf{x}_2$ *with* $\mathbf{l}_1$*, and* $\mathbf{x}'_2$ *the intersection of* $\mathbf{x}_1 \times \mathbf{x}_2$ *with* $\mathbf{l}_2$*. Then*

$$\cos^2(\theta) = \frac{|\mathbf{x}_2\mathbf{x}'_1|\,|\mathbf{x}_1\mathbf{x}'_2|}{|\mathbf{x}_1\mathbf{x}'_1|\,|\mathbf{x}_2\mathbf{x}'_2|}\,, \tag{5}$$

56

Finding the calibrating conic.

Assumptions:

1. Pixels are square
2. Principal point at the centre of the image.
3. All that remains to find is the focal length.

The Calibrating Conic

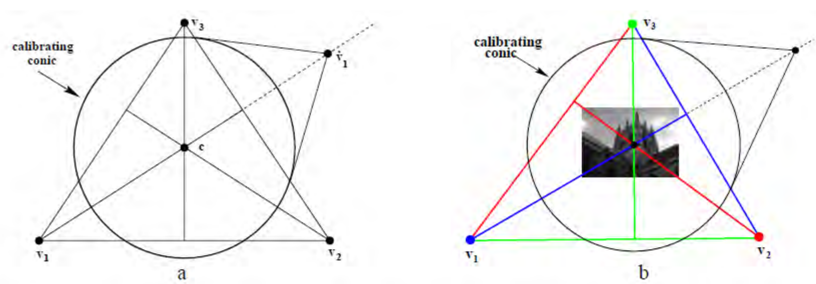
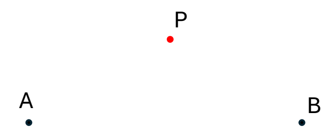
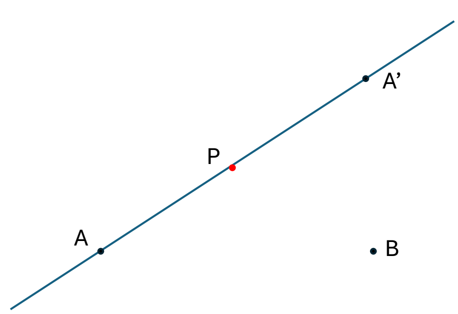


Fig. 8.29. *The calibrating conic computed from three orthogonal vanishing points. (a) The geometric construction. (b) The calibrating conic for the image of figure 8.22.*



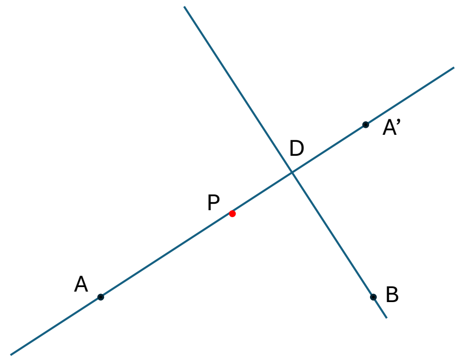
Task: Find the calibrating conic. C is the principal point, and rays through A and B are perpendicular.

59



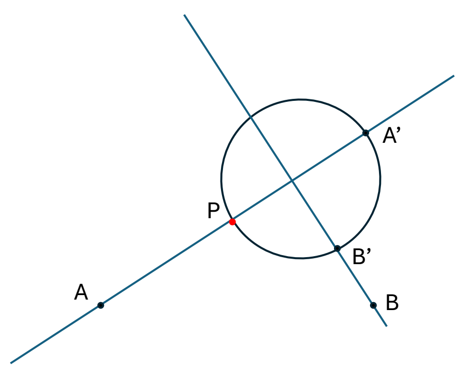
Step 1. Find the reflection of A through principal point P

60



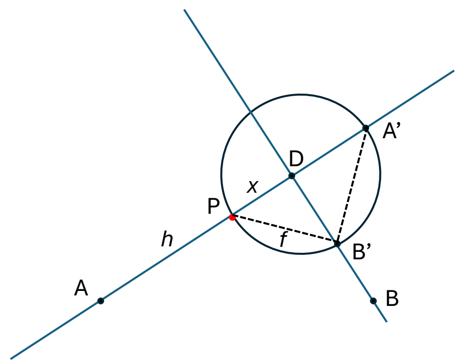
Line BD must be the polar of A' (since A is perpendicular to B.

61



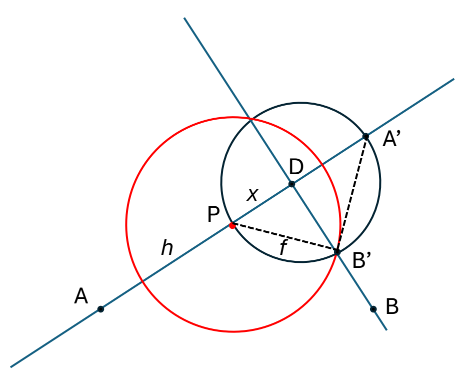
Draw circle with diameter A'P and intersect perpendicular a B'

62



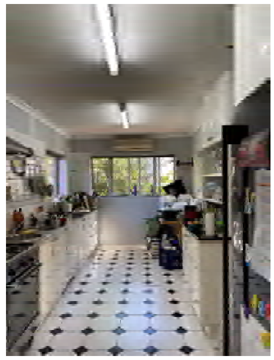
Angle A'BP must be right angle, so A'B' is a tangent to a circle centre P through B'.

63

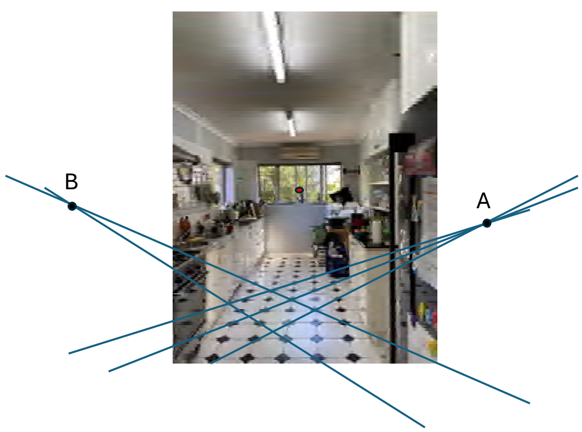


Red circle is the calibrating conic.

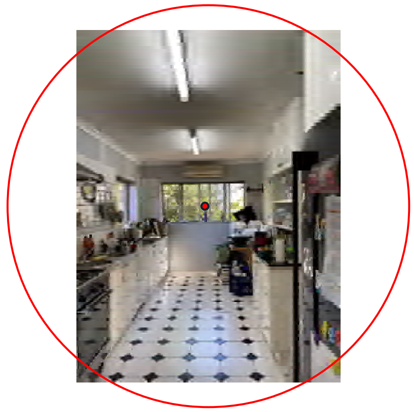
64



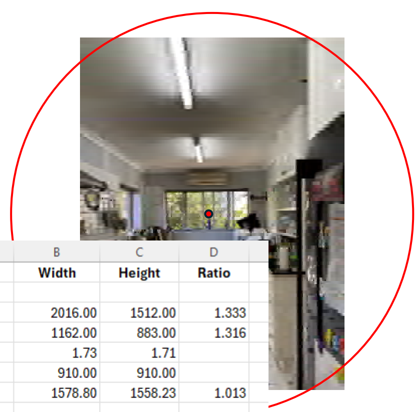
65



66



67



	A	B	C	D
1		Width	Height	Ratio
2	Image			
3	W/H (pixels)	2016.00	1512.00	1.333
4	W/H (mm)	1162.00	883.00	1.316
5	Pixels / mm	1.73	1.71	
6	Camera distance (mm)	910.00	910.00	
7	f (pixels)	1578.80	1558.23	1.013
8				
9	Tan (theta_h)	0.638	0.485	
10	theta/2	32.557	25.881	
11				
12	Width start (mm)	66.00	85.00	
13	width end (mm)	1228.00	968.00	

68

69

70

71

Which is closer

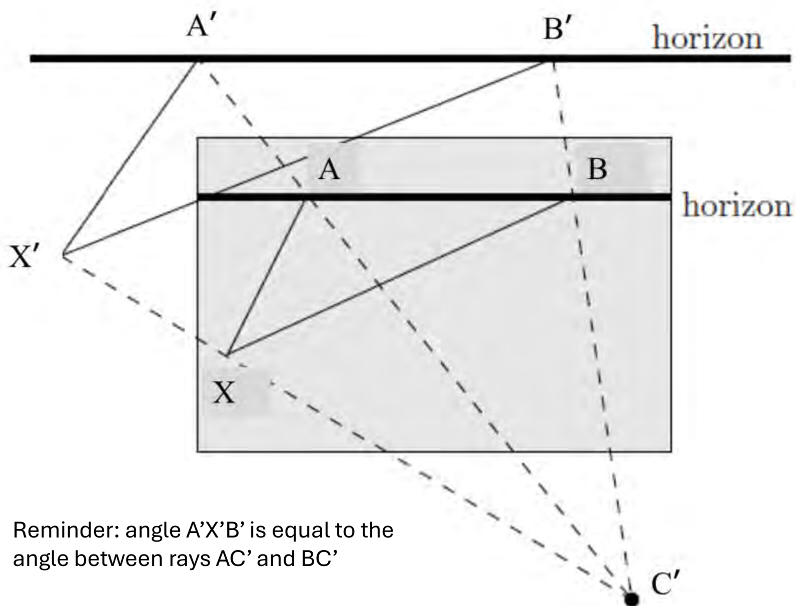
72

Which is closer

73

Conformal point

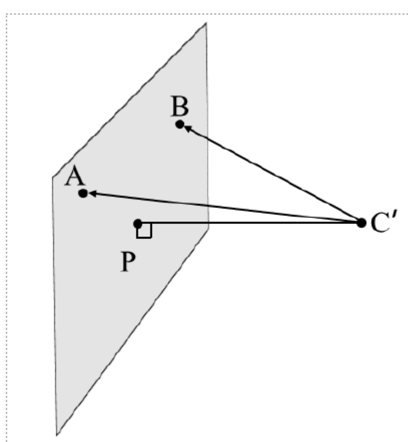
74

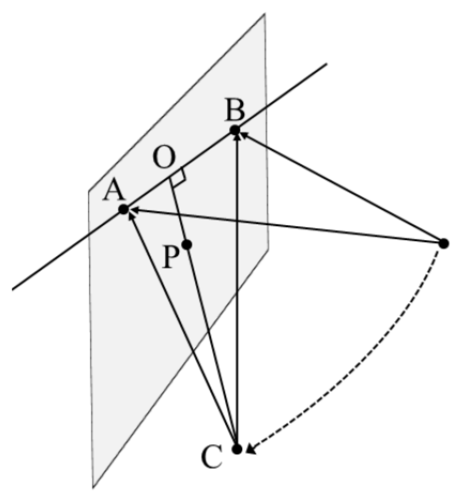


Reminder: angle A'X'B' is equal to the angle between rays AC' and BC'

But how do we find the angle between rays?

75

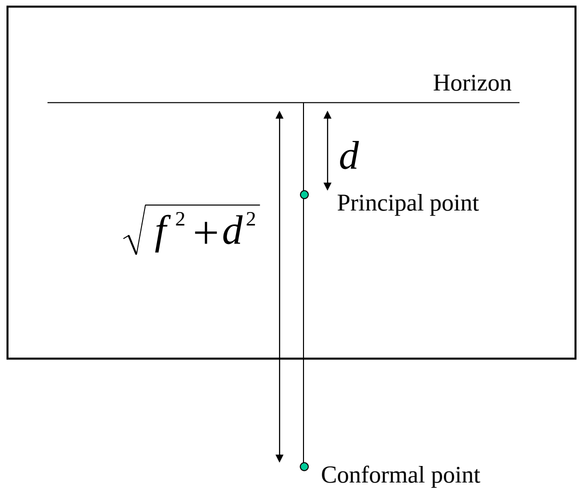




How to find the angle between rays through A and B.

- Rotate C' about the axis AB to point C in the image.
- Then angle ACE = angle AC'E.

76

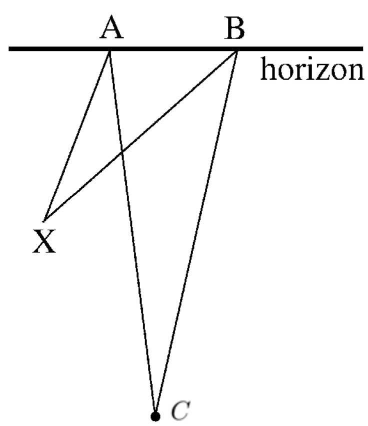


Conformal point lies a distance $(f^2 + d^2)^{1/2}$ from the horizon.

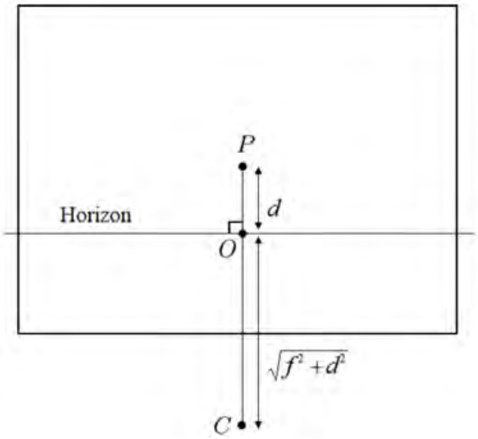
77

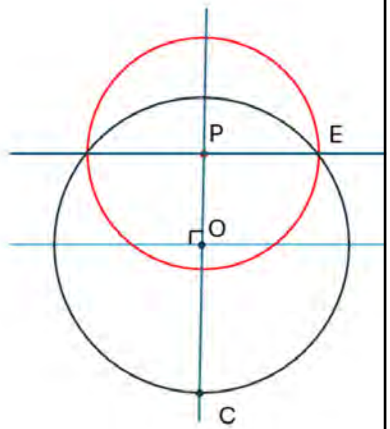
Angle measurement

- Extend lines to the horizon.
- Connect back to the conformal point



78

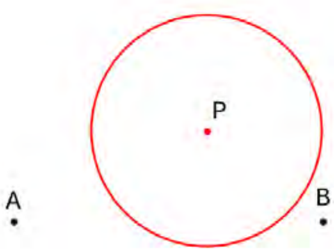


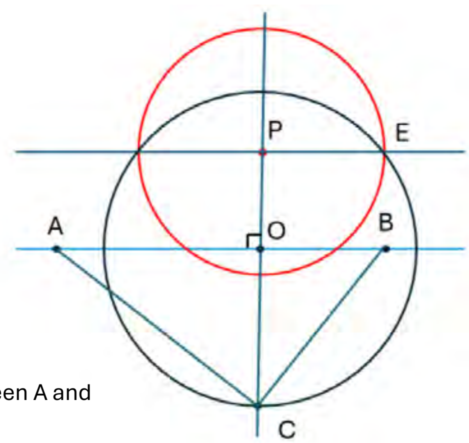


Conformal point for a plane lies on perpendicular to horizon, at the shown distance.

Geometric construction for finding conformal point. C is the point such that |OC| = |OE|

79



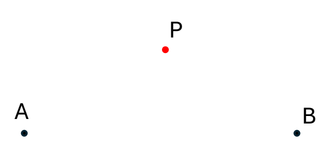


- Measuring angles. Find the angle between A and B, knowing the calibrating conic.
- Angle is equal to angle ACB.
- If A and B lie on a horizon line, then conformal point only has to be computed once for the given horizon.

80

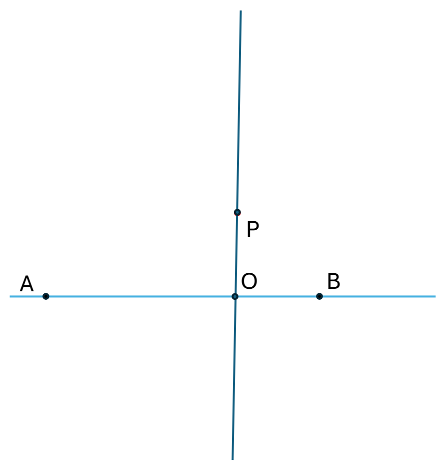
Finding the calibrating conic using the conformal point method.

81



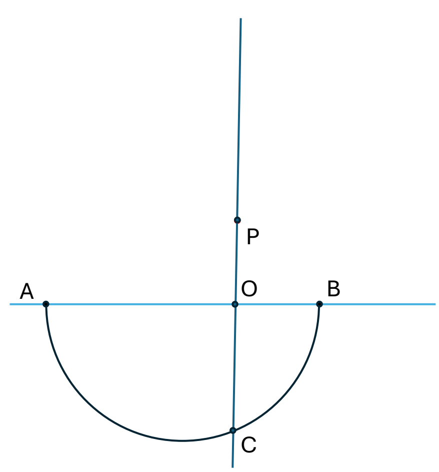
Task: Find the calibrating conic. C is the principal point, and rays through A and B are perpendicular.

82



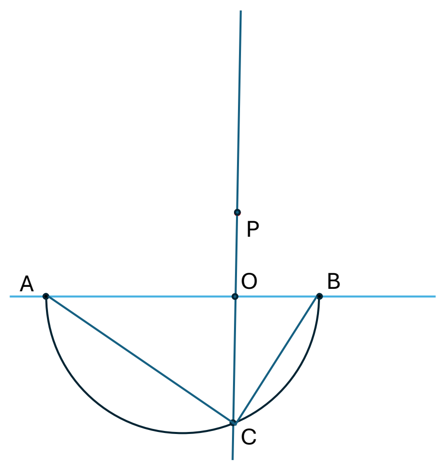
Step 1: Join AB and take perpendicular through P

83



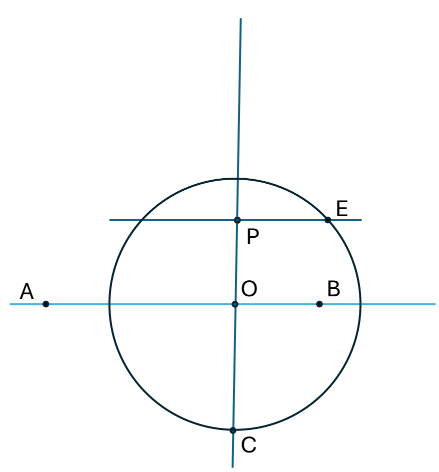
Step 2: Draw (semi) circle with radius AB and find intersection C with perpendicu
Then C is the conformal point for line AB.

84



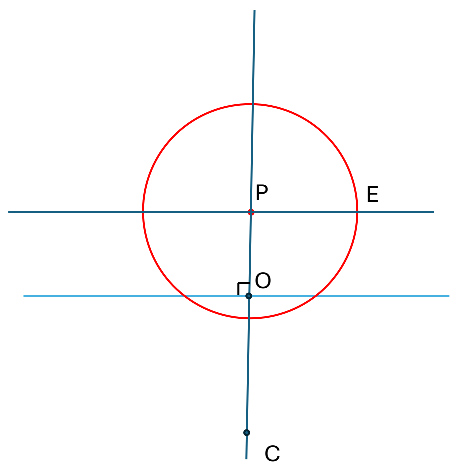
... because angle ACB is a right angle.

85



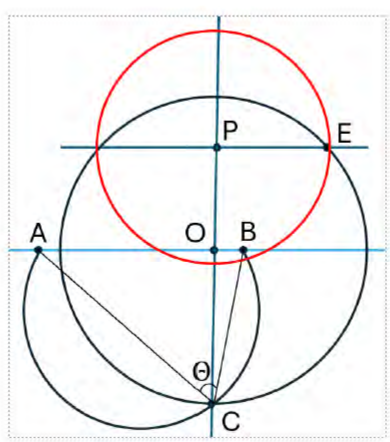
Step 3: Intersect circle centre O and radius |OC| with line through P parallel to AB.

86



Step 4: Calibrating conic is the circle with radius PE.

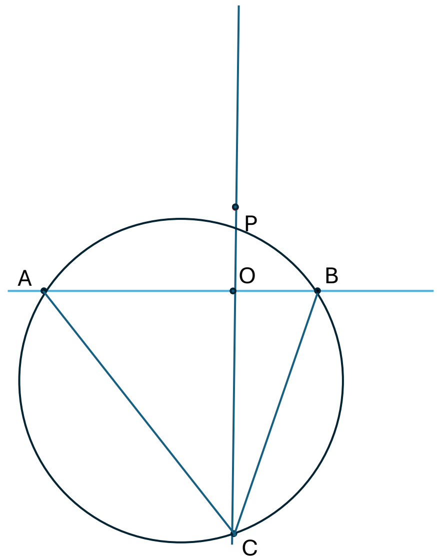
87



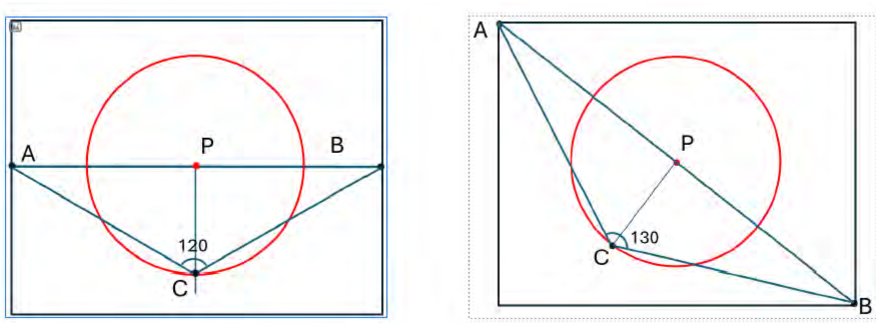
Conformal point method generalizes to observation of any known angle (e.g. 45 degrees) seen in the image.

88

Only difference: Draw a given-angle circle through A and B, i.e. such that angle ACB is the desired angle



89



Finding the field of view angle, that is, the angle between rays A and B. Conformal point lies on the calibrating conic.

90

Methods of computing the horizon

- Interactively drawing it (once only)
- From constructing two vanishing points
- From planar motion : line passing through epipoles.
- Length ratios on lines in an image

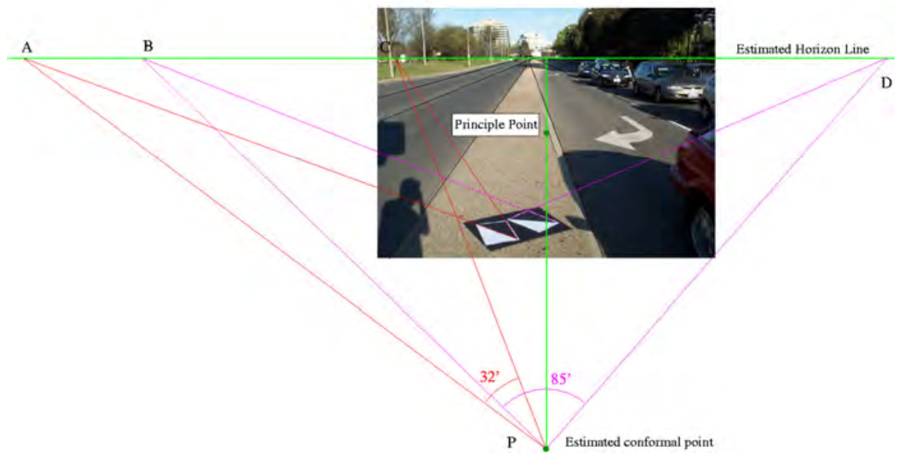
91

Computing the conformal point

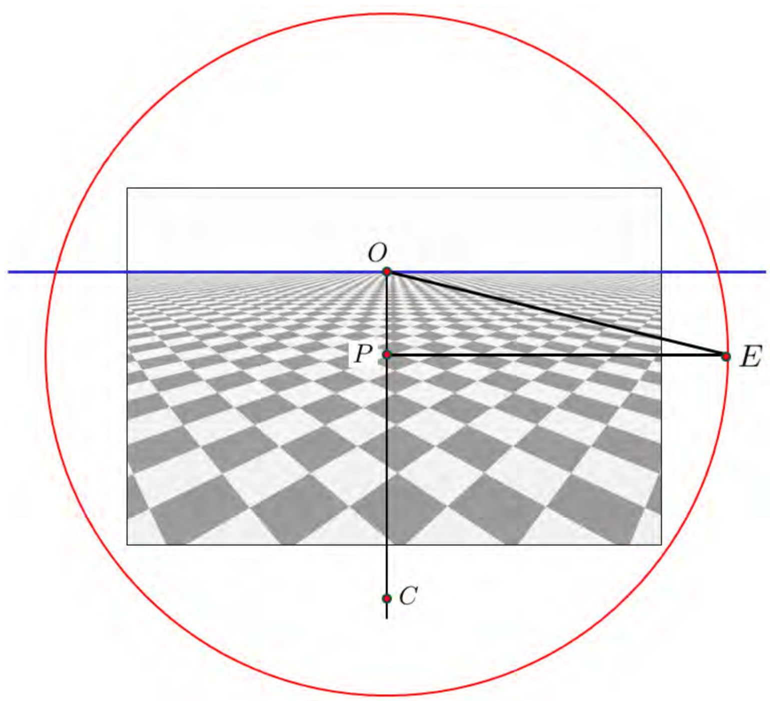
- Directly from known focal length and principal point.
- From known principal point and an observed known angle
- From two known observed angles
- From three equal unknown angles.
- Planar motion : from the motion of three points from one frame to the next.

92

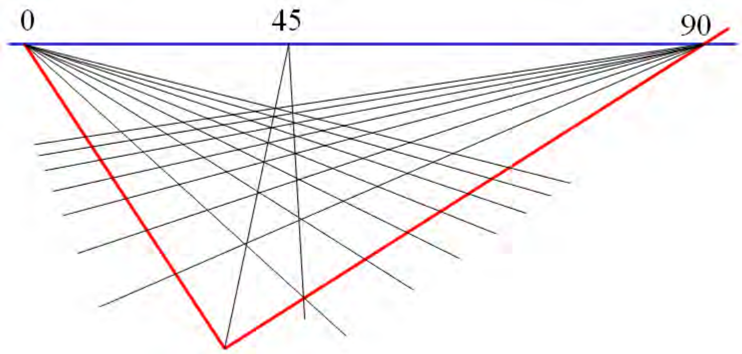
Example



93

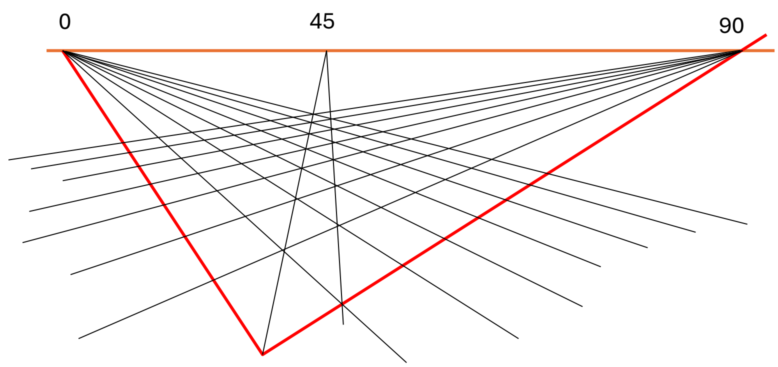


94



95

Perspective and Art-history



Drawing perspectively correct pavement grids.

96

Application : Sun direction

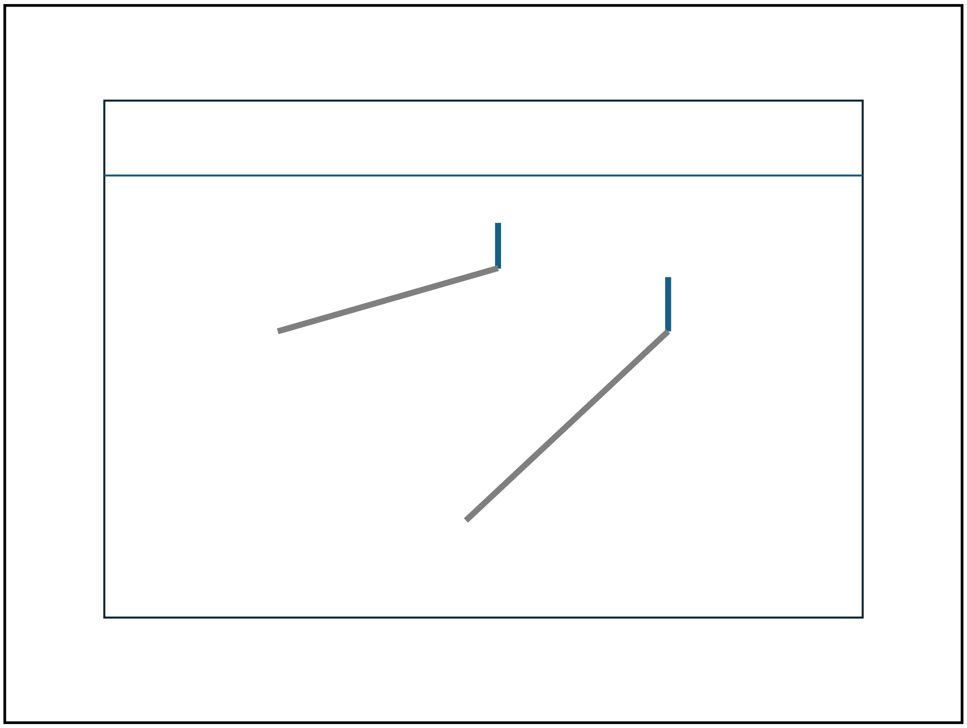
- Measure angle between shadows and building lines

97

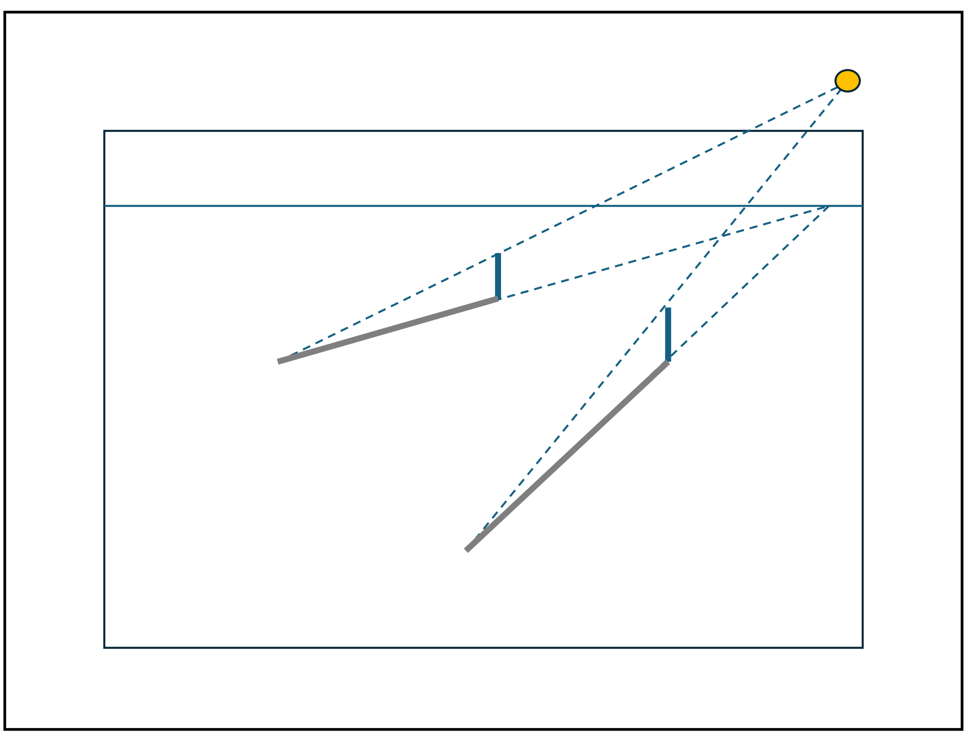
Application : Sun direction

- Measure angle between shadows and building lines

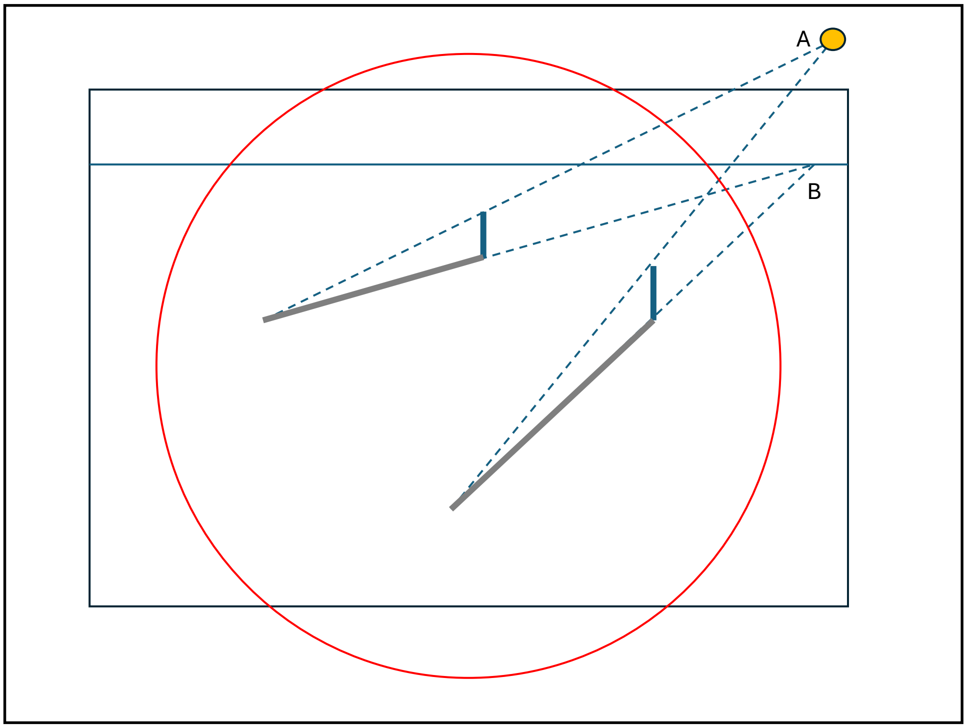
98



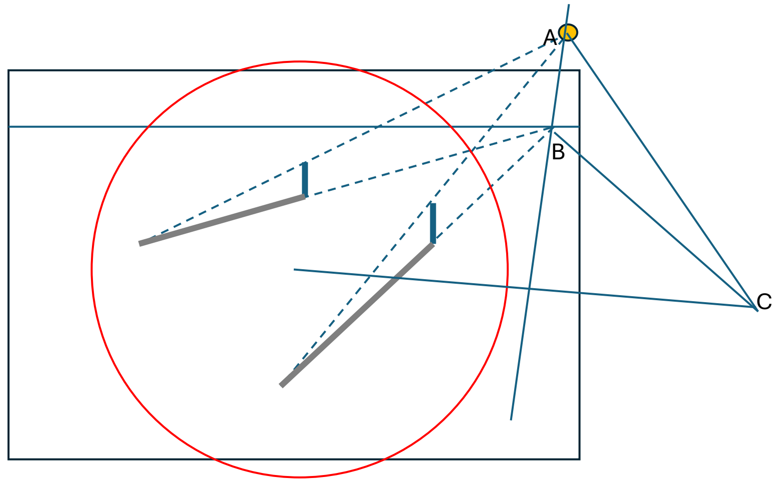
99



100

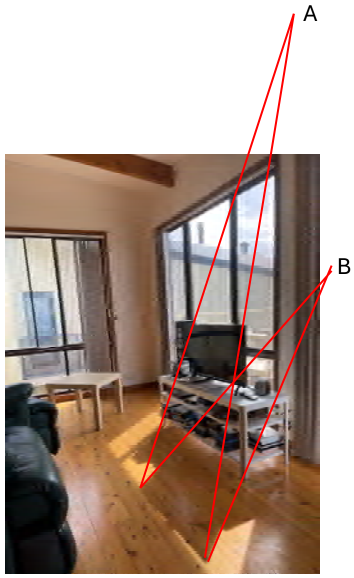


101



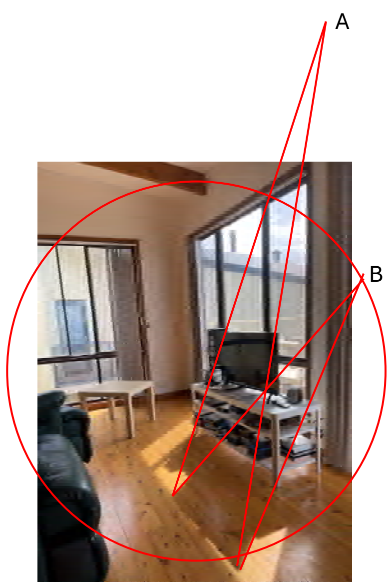
C is the conformal point
Angle ACB is the sun elevation.

102



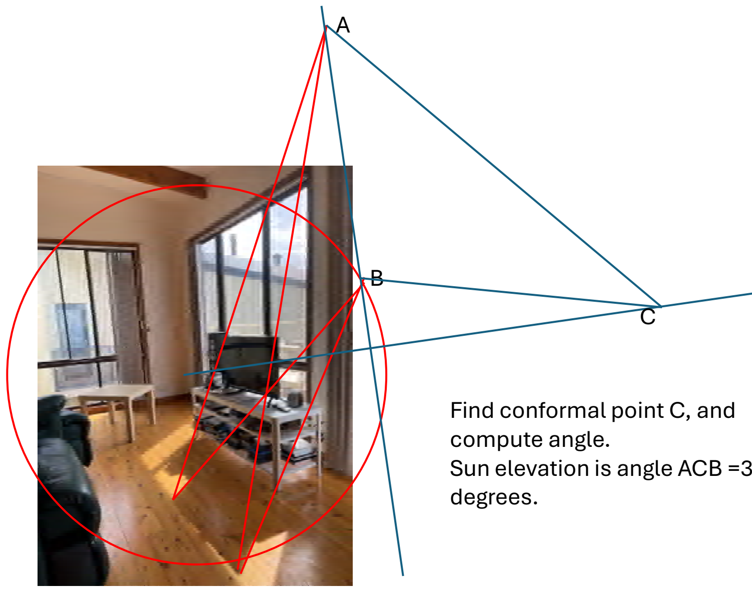
Construct sun position, and point on horizon below sun

103



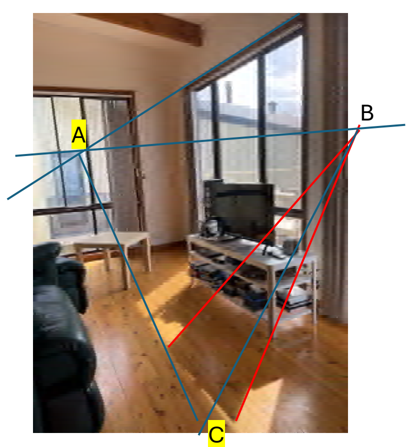
Overlay calibrating conic

104



Find conformal point C, and compute angle.
Sun elevation is angle ACB =35 degrees.

105



Computing bearing of the sun, with respect to orientation of my room.

C is the conformal point for the ground plane.

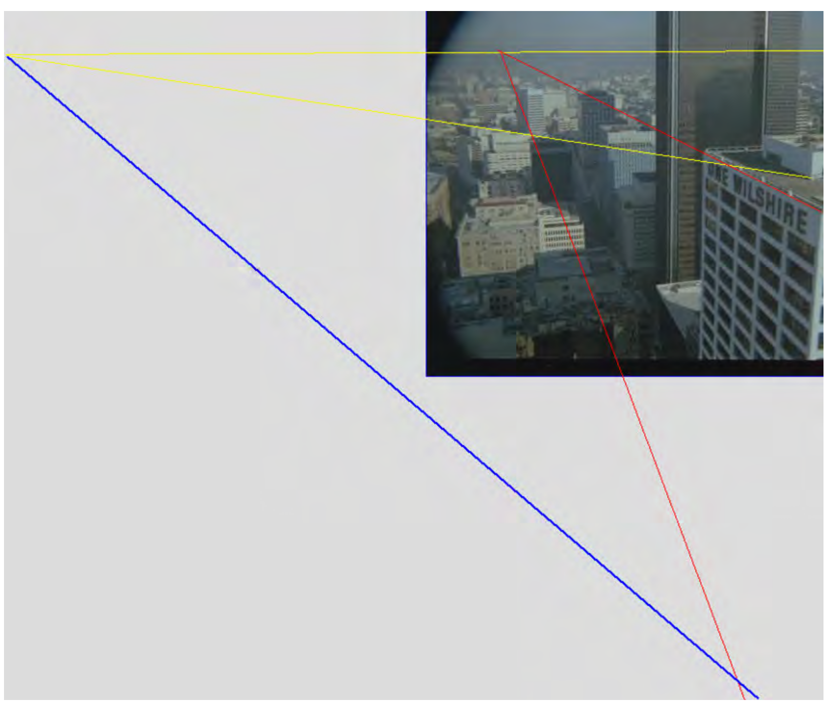
AB is the horizon line for ground plane.

Sun bearing, with respect to room is angle ACB.

106

- Picture was taken at 9:59 am on Feb 4.
- Consulting sun tables:
  - My house is at 35 degrees south, 135 degrees east.
  - Living room faces at a bearing of 127 degrees (based on bearing to sun)

107



108

Art history

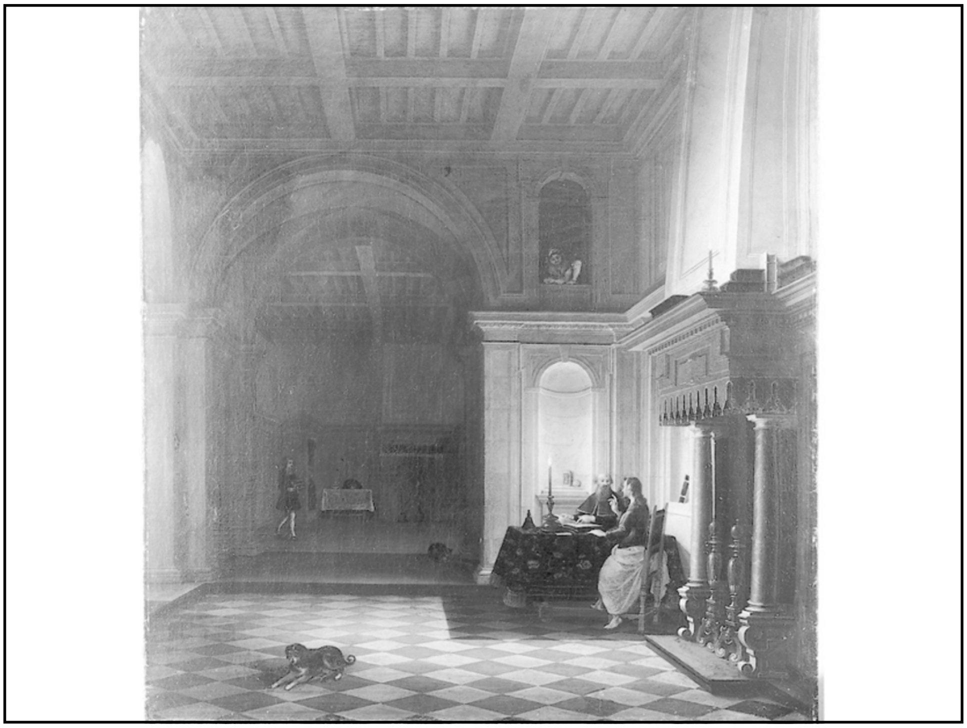
109

Rules for perspective enunciated by Brunelleschi

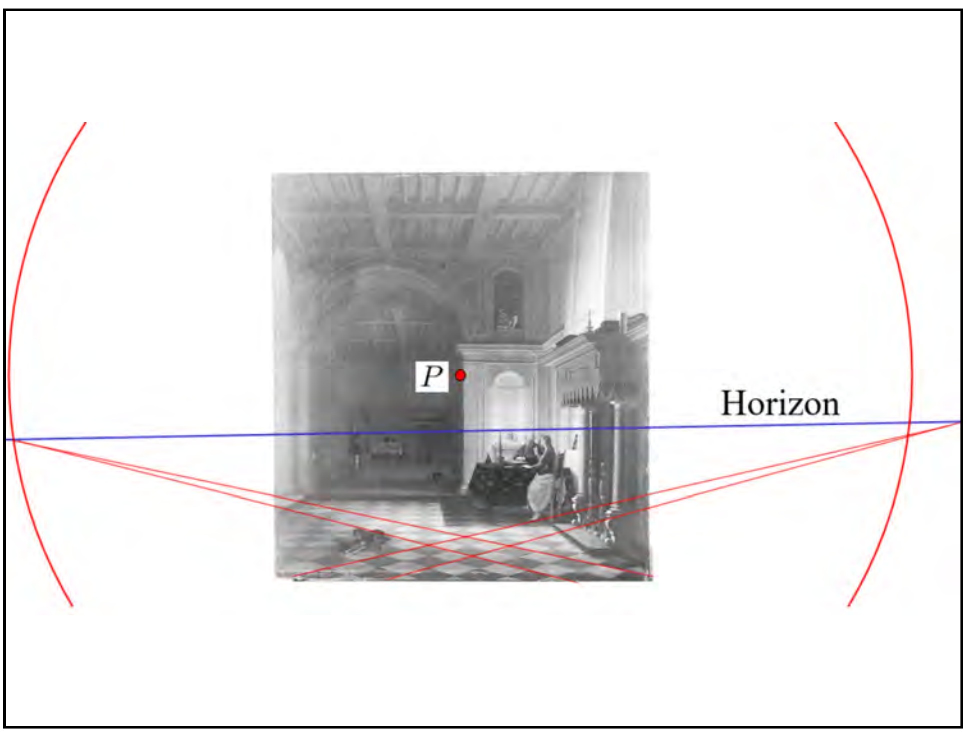
Method for drawing perspectively correct planes given by Leon Battista Alberti, “De Pictura” (1435)

http://www.mega.it/eng/egui/pers/lbalber.htm

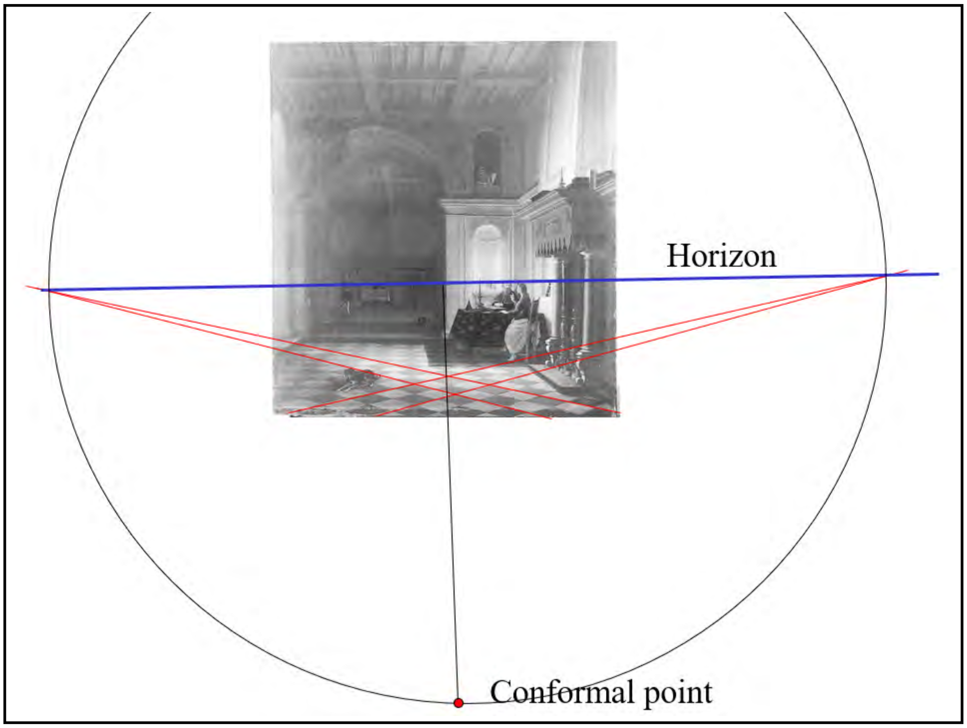
110



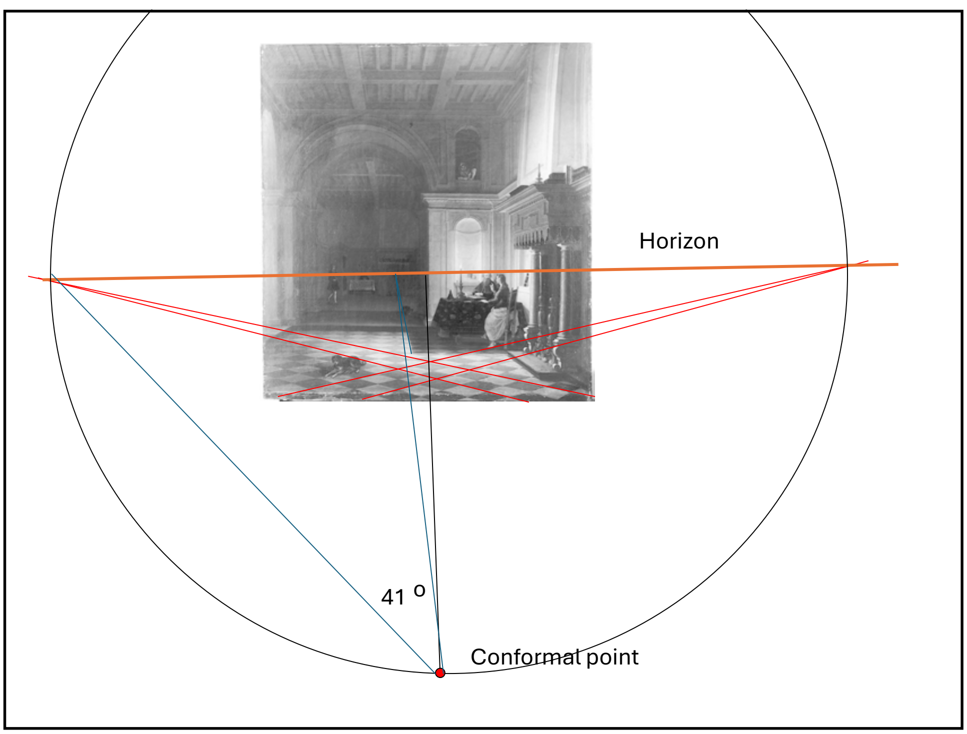
111



112



113



114

The field of view may be measured as a half angle of 24.2° vertically and 22.5° horizontally. Since the principal point lies above the horizon, the view is in an upward tilted direction. The angle of upward tilt is about 6.7°, as estimated by the method described in fig 18. Finally, the view direction is at an angle of about 5.4° towards the right with respect to the orientation of the room, measured by the vanishing point of the floor tiles, the rafters in the roof and the upper edge of the mantle-piece above the fireplace. All of these lines converge at points on the horizon.
(By a rough measure, taking into account a subjective feeling for the precision of the construction and measurements, an accuracy of around 2% or 3% error in these estimates seems justified.)

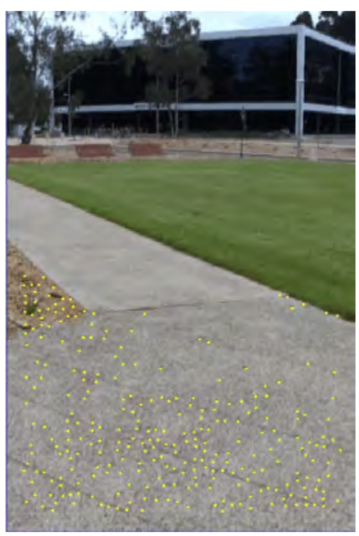
115

Application: Odometry

116

Application:
- Robot moving in a plane
- Camera fixed with respect to robot
- Thus: camera rotation is about axis perpendicular to plane.
- The horizon and conformal point are fixed for the whole motion, so need only be computed once.
- Camera (and hence platform) rotation can be computed by tracking just two points, say x, y
- Angle between XY and X'Y' gives the angle of rotation of the platform (independent of translation)
- Allows for very efficient robust rotation (RANSAC with model computable from only two points.

117

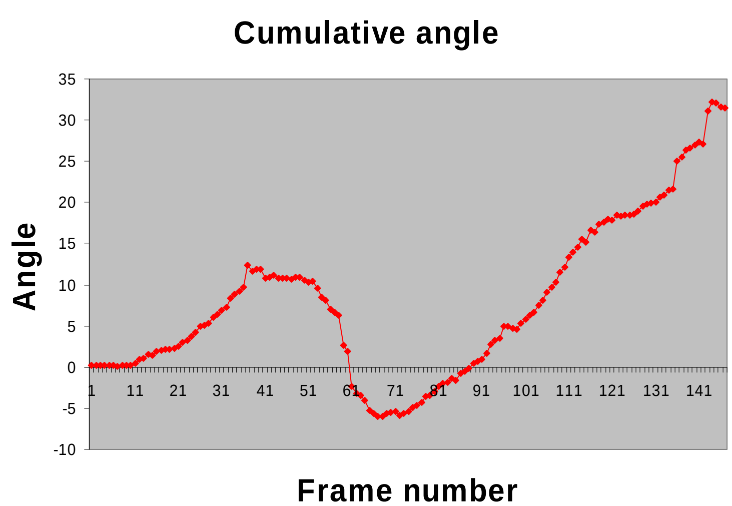


Tracked points

118

- Horizon and conformal point easily computed from two views by computing the homography from point matches.
- Only do once.
- Chat-gpt knows how to do this.

119



120